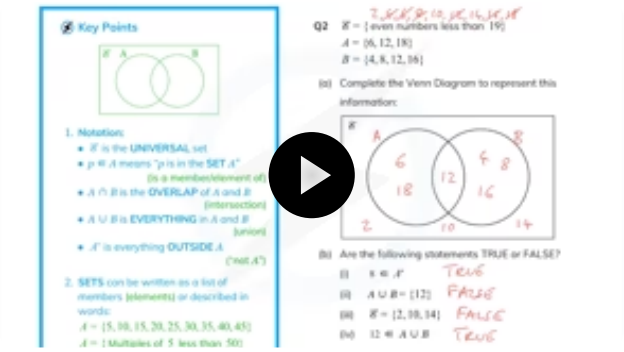
Set Notation & Venn Diagrams

Exam code: 0607

Download PDF

Written by: Mark Curtis
Reviewed by: Dan Finlay
Updated on 27 January 2026



Did this video help you? Yes No

Set notation

What is a set?

- A set is a **collection of elements**
  - Elements could be anything
    - Numbers, letters, coordinates, ...
- You could describe a set by writing its elements inside **curly brackets** {}
  - {1, 2, 3, 6} , is the set of factors of 6
- If the set of elements **follow a rule** then you can write this using a **colon inside the curly brackets** {... : ...}
  - The bit before the colon is the type of element
  - The bit after the colon is the rule
    - {x is a positive integer : $x^2 < 30$} is the set of positive integers which, when squared, are less than 30
    - This is equal to {1, 2, 3, 4, 5}
  - The colon is often read as '**such that**'
  - If **no type** is specified, x can take **any value** (fractions, decimals, irrationals, ...)
    - {x: $x^2 < 30$} means any value whose square is less than 30
  - { (x, y) : $y = mx + c$ } would mean the coordinates (x, y) where $y = mx + c$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

        - $\{x \mid x^2 < 30\}$

What do I need to know about set notation?

- $\mathscr{E}$ is the **universal** set (the set of **everything**)
  - For example, if we are only interested in factors of 24 then $\mathscr{E} = \{1, 2, 3, 4, 6, 8, 12, 24\}$
  - You may see alternative notations used for $\mathscr{E}$
    - U is a common alternative (different to $\cup$ for union!)
    - S or the Greek letter ξ (xi) may also be seen

- We use **upper case** letters to represent **sets** (A, B, C, ...) and **lower case** letters to represent **elements** (a, b, c, ...)
- n(A) is the **number of elements** in set A
  - For example, if $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 4, 9\}$, $B = \{1, 2, 3, 4, 5, 6\}$
    n(A) = 3, n(B) = 6

- $\in$ means "**is an element of**"
  - And $\notin$ means "is not an element of"
  - E.g. $3 \in \{1,2,3\}$ and $4 \notin \{1,2,3\}$

- $\varnothing$ is the **empty set**
  - This is the set which does not contain any elements

- $A \subseteq B$ means "A is a **subset** of B"
  - Set A is a subset of set B if all the elements of A are also present in B
    - E.g. If $A = \{1,2,3\}$ and $B = \{0, 1, 2, 3, 4\}$, then $A \subseteq B$

  - $A \subsetneq B$ means "A is not a subset of B"

- $A \cap B$ means the **intersection** of A and B (the **overlap** of A **and** B)
  - This is the set of elements that are in **both** set A and set B
    - E.g. If $A = \{1, 2, 3, 4, 5\}$ and $B = \{-1, 1, 4, 7, 8\}$, then $A \cap B = \{1, 4\}$

- $A \cup B$ means the **union** of A and B (**everything** in A **or** B **or** both)
  - This is the set of elements that are in **at least one** of the sets
  - This **includes** elements in both sets (in the intersection)
    - E.g. If $A = \{5, 6, 7, 8\}$ and $B = \{3, 7, 11\}$, then $A \cup B = \{3, 5, 6, 7, 8, 11\}$

- A' means the **complement of *A***
  - It is the **set** of all elements in the universal set $\mathscr{E}$ that are **not in *A***
    - E.g. If $\mathscr{E} = \{1, 2, 3, 4, 5\}$ and $A = \{1, 3\}$, $A' = \{2, 4, 5\}$

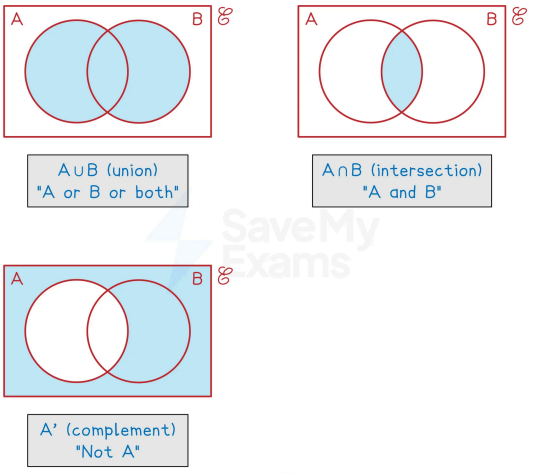
Sets & Venn diagrams

What is a Venn diagram?

- a **rectangle** representing the **universal set ($\mathscr{E}$)**
- a **circle** for each **set**
  - Circles may or may not overlap depending on which **elements** are shared between **sets**

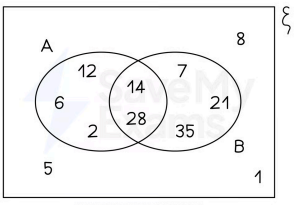
What do the different regions mean on a Venn diagram?

- $A \cup B$ is represented by the regions that **are in** A or B or both
- $A \cap B$ is represented by the region where the A and B circles **overlap**



Worked Example

Two sets A and B are shown in the Venn diagram.



(a) Write down n(A).

Answer:

The elements of A are anything inside the A circle

$$A = \{2, 6, 12, 14, 28\}$$

n(A) means the number of elements in A
There are 5 elements in A

n(A) = 5

(b) Use set notation to complete the sentence {14, 28} = ...

Answer:

{14, 28} = A ∩ B

(c) Write down the elements that are in set $A \cup B$.

Answer:

$A = \{2, 6, 12, 14, 28\}$ and $B = \{7, 14, 21, 28, 35\}$
$A \cup B$ is the set of elements that are in at least one of the sets

For elements in both, only write them out once

$A \cup B = \{2, 6, 7, 12, 14, 21, 28, 35\}$

(d) Jamie states that $A' \cap B' = \varnothing$.

Explain if this statement is correct or not.

Answer:

The statement means Not A and Not B is equal to an empty set
i.e. There are no elements that are Not A and Not B

However, there are 3 elements that are neither A nor B

There are 3 elements that aren't in A or B
$A' \cap B' = \{1, 5, 8\}$

Therefore the statement is incorrect

Worked Example

Consider the following sets.

$$\mathscr{E} = \{\text{letters of the alphabet}\}$$
$$T = \{t, e, a, m\}$$
$$I = \{i\}$$
$$M = \{m, e\}$$

Fill in the blanks with the appropriate set.

(i) ... $\nsubseteq T$

(ii) ... $\subseteq$ ...

Answer:

The symbol $\subseteq$ means "is a subset of", so all of the elements of one feature in the other
The symbol $\nsubseteq$ means "is not a subset of"

The element in I is not in T

$$I \nsubseteq T$$

All the elements of M are also in T

$$M \subseteq T$$

Teacher only

Create exam-aligned tests
Tailor format & difficulty
Quizzes or homework in minutes
Try Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.